Typical errors in students' solutions:

Question 1

a) Almost all got the field lines correct, but only a few students could articulate all the symmetry properties.
b) The most typical, but minor error, was to plot the fields only for positive x. Note also that the problem could have been solved using differential form of Maxwell's equations. Several students attempted it, but only one succeeded.
c) The most typical errors were: incorrect sign of the final expression, incorrect dimensions of the final expression, and errors in integration. Some students derived the field inside the slab in b) using the total height of cylinder, but forgot to divide it by two when calculating voltage. Some answers incorrectly stated that the voltage outside the slab was zero because the field was homogeneous. Note that there was no penalty for using incorrect fields if such were obtained in b); several students incorrectly derived the fields in b) but still got maximum marks for c).

Question 2

a) Most students had no problems here
b) Most students had no problems here. Several answers mistakenly tried to use a closed surface when applying Ampere's law.
c) The most common mistake was inability to formulate the integral. A common subset of this mistake was to assume r=const in Biot-Savart's law and take it out of the integral. Another typical erroneous strategy was to fiddle with expressions to force the answer agree with the result of b). A possible alternative to the model answer is to formulate the integral in terms of an angle. Several answers attempted it but only a few succeeded.

Question 3

a) The most common recurring mistakes were to forget about H or to say that H=const in an ideal transformer, whereas the correct answer is H=0.
b) There were two common mistakes: first, not being able to derive the expression for the currents, typically, when forgetting about H in a); second, postulating the expressions from memory instead of deriving them
c) The most common mistakes were being unable to assign the meaning of the components or mistaking one for another, Xm and Ri causing the most frequent problems. Some answers discusses the circuit tests, which was not asked.